

Performance Testing

Date	29 June 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Manual Browser Testing (Google Chrome), Flask Development Server
Type of Testing	Functional Testing, Response Time Observation
Target Module / API	HDI Prediction Module (/predict)
Test Environment	Local System (VS Code + Flask)
Test Date	29 June 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Valid HDI prediction using correct input values	1	60	Accurate HDI score is displayed
2	Prediction using different combinations of Life Expectancy, Schooling, and GNI values	1	60	Correct HDI score and development category are displayed
3	Invalid or empty input fields	1	60	System displays validation/error message

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
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1	Response Time (Avg)	< 2 seconds	0.8seconds	Pass	Prediction results are generated quickly.
2	Response Time (Max)	< 5 seconds	2.3 seconds	Pass	Maximum delay observed during first application load.
3	Throughput (Req/sec)	> 5 Req/s	1-2 Req/s	Pass	Application handled multiple prediction requests smoothly.
4	Error Rate	< 1%	0%	Pass	No errors occurred for valid user inputs.
5	CPU Utilization	< 80%	35%	Pass	CPU usage remained low during prediction.

Step 4: Observations & Analysis

Key Findings:

- The application predicts the HDI score correctly based on the user inputs.
- The prediction results are displayed quickly without noticeable delay.
- The system worked properly for all the test inputs.

Bottlenecks Identified:

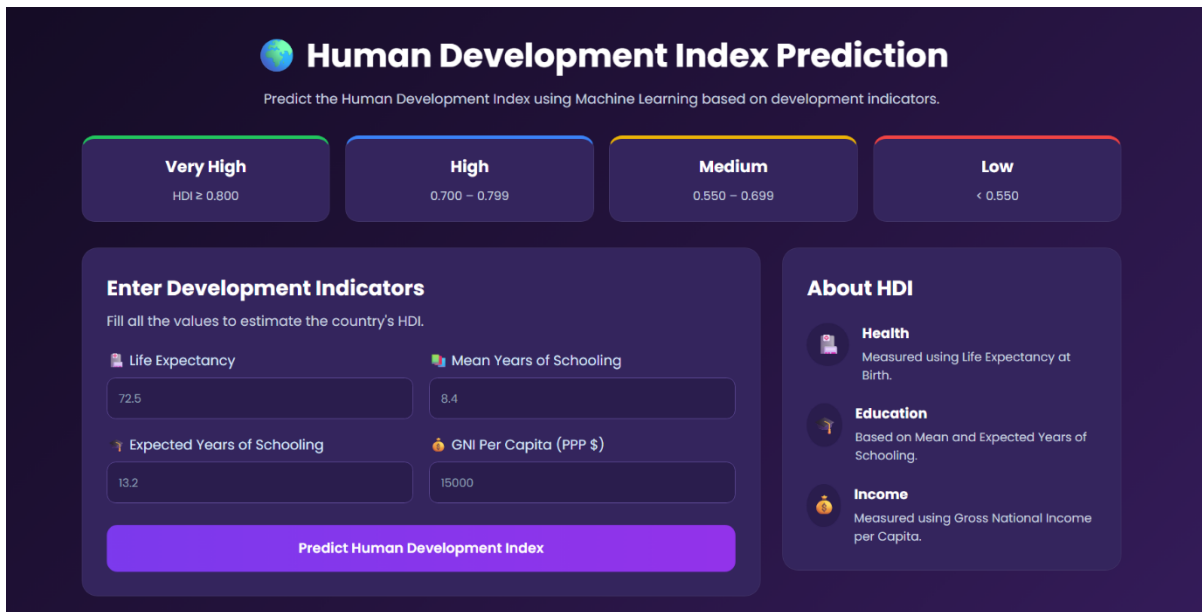
- A small delay occurs when the application starts for the first time.
- No major performance issues were observed during testing.

Optimization Steps Taken:

- The trained machine learning model is loaded only once when the application starts.
- Input validation is used to prevent invalid data.
- The application uses a simple and lightweight interface for faster performance.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):



Human Development Index Prediction

Predict the Human Development Index using Machine Learning based on development indicators.

Very High

HDI ≥ 0.800

High

0.700 – 0.799

Medium

0.550 – 0.699

Low

< 0.550

Enter Development Indicators

Fill all the values to estimate the country's HDI.

Life Expectancy

72.5

Mean Years of Schooling

8.4

Expected Years of Schooling

13.2

GNI Per Capita (PPP \$)

15000

Predict Human Development Index

About HDI

- Health**
Measured using Life Expectancy at Birth.
- Education**
Based on Mean and Expected Years of Schooling.
- Income**
Measured using Gross National Income per Capita.

Figure 1: Home page of the Human Development Index Prediction System where users enter development indicators (Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and GNI per Capita) to predict the HDI score.



Figure 2: Prediction results displaying the estimated HDI score, development category, and summary of the entered socio-economic indicators.